Global Valuation

Cross-Country Valuation Multiples: Data and First Results

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Outline

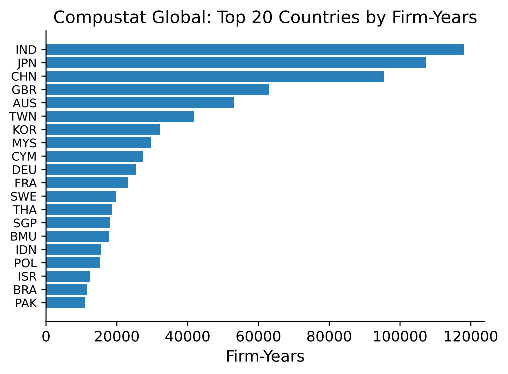
Data

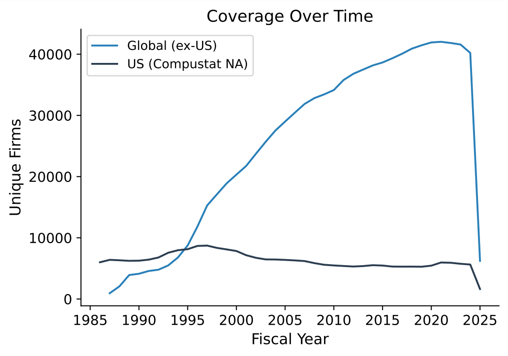
Datasets

File	Size	Rows	Cols	Description
compustat_global	2.3 GB	981K	468	Global annual fundamentals (124 countries)
compustat_america	0.8 GB	254K	984	NA fundamentals + CRSP link
returns_global	1.9 GB	10.4M	25	Global monthly prices
returns_america	0.9 GB	3.6M	55	CRSP US monthly returns

- Compustat Global: fiscal years 1987–2025, 124 countries
- Compustat America: fiscal years 1986–2025, CRSP-linked
- Combined regression panel: **749,527 firm-years**, 72,914 firms, 81 countries

Coverage: Countries and Time





India, Japan, China dominate non-US coverage. Panel peaks at ∼41K firms (2022).

Panel Construction

- **Market-to-book**: $\log(M/B) = \log(\text{Market Cap}/\text{Book Equity})$
  - US: `mkvalt` from Compustat NA; Global: `cshoi` $\times$ `prccm`
- **ROA**: Net Income / Total Assets **Size**: log(Total Assets)
- **Lag earnings growth**: $(E_{t-1} - E_{t-2})/|E_{t-2}|$ **Industry**: SIC-2
- Winsorize M/B at 1st/99th percentiles; filter country-years with $\geq$ 30 obs

Variable	N	Mean	P10	Median	P90
$\log(M/B)$	749,527	0.46	−0.77	0.39	1.76
ROA	749,093	−0.02	−0.15	0.02	0.11
log(Assets)	749,527	7.28	3.26	7.17	11.49

Cross-Country Regressions

Regression Specifications

Market-to-Book:

$$\log(M/B)_{i,t} = \alpha + \beta_1 \mathrm{ROA}_{i,t} + \beta_2 \log(\mathrm{Assets})_{i,t} + \beta_3 \Delta E_{i,t-1} + \gamma_c + \delta_j + \tau_t + \varepsilon_{i,t} \tag{Spec 1}$$

$$\log(M/B)_{i,t} = \alpha + \beta_1 \mathrm{ROA}_{i,t} + \beta_2 \log(\mathrm{Assets})_{i,t} + \beta_3 \Delta E_{i,t-1} + \delta_j + \gamma_{c,t} + \varepsilon_{i,t} \tag{Spec 2}$$

ROA:

$$\mathrm{ROA}_{i,t} = \alpha + \beta_1 \log(\mathrm{Assets})_{i,t} + \gamma_c + \delta_j + \tau_t + \varepsilon_{i,t}$$

γ_c = country FE, δ_j = industry FE (SIC-2), τ_t = year FE, $\gamma_{c,t}$ = country × year FE.
Standard errors clustered at the country level.

M/B Regression Results

	Spec 1	**Spec 2**
	C + Ind + Year FE	C×Year + Ind FE
ROA	−0.070	−0.102
	(0.133)	(0.129)
log(Assets)	−0.030	−0.032
	(0.024)	(0.024)
Lag Earn. Growth	0.006***	0.005***
	(0.002)	(0.002)
N / R^2	607,209 / 0.185	607,209 / 0.211
C / Ind / Year FE	Y / Y / Y	– / Y / –
C × Year FE	–	Yes

*** $p < 0.01$. Clustered SEs at country level.

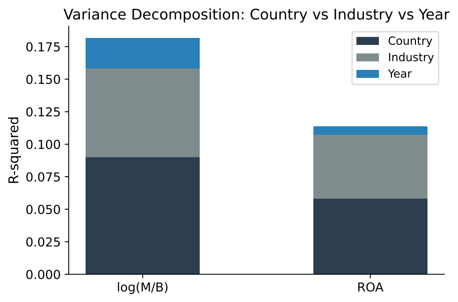
Key Takeaway: ROA Does Not Explain M/B

- ROA coefficient is **insignificant** in both specifications
- After country, industry, and year controls, current profitability does not predict M/B
- Only **lag earnings growth** is significant (positive, as expected)
- Size effect negative but insignificant with clustered SEs
- **Interpretation**: cross-country M/B differences reflect *discount rates* and *growth expectations*, not current profitability

R^2 **comparison:** Spec 1 $= 0.185$, Spec 2 $= 0.211$ — country$\times$year interactions gain 2.6 pp, suggesting time-varying country effects matter.

Variance Decomposition

What Drives Valuation? Country vs Industry vs Year



Shapley R^2 decomposition:

log(M/B):

- Country: **49.6%** of R^2
- Industry: 37.7%
- Year: 12.7%

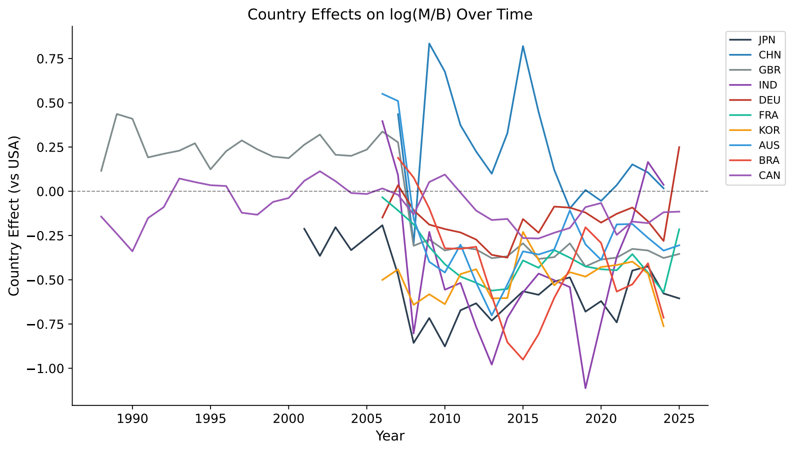
ROA:

- Country: **51.2%** of R^2
- Industry: 43.2%
- Year: 5.6%

Country dominates both valuation and profitability.

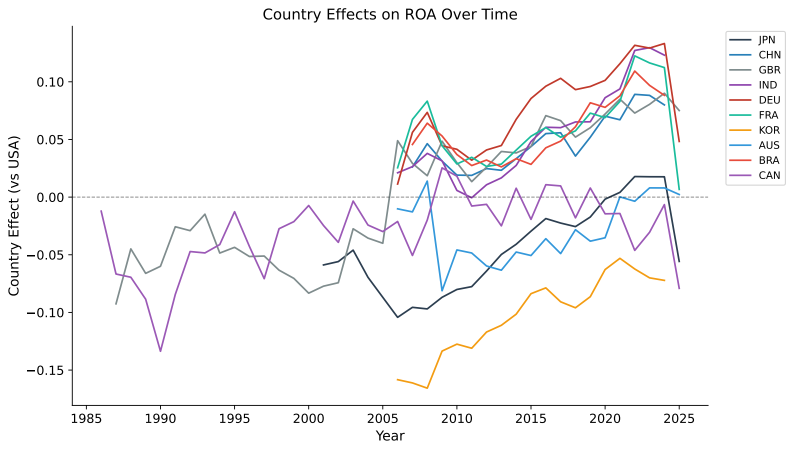
Country Effects Over Time

Country Effects on M/B (vs USA)



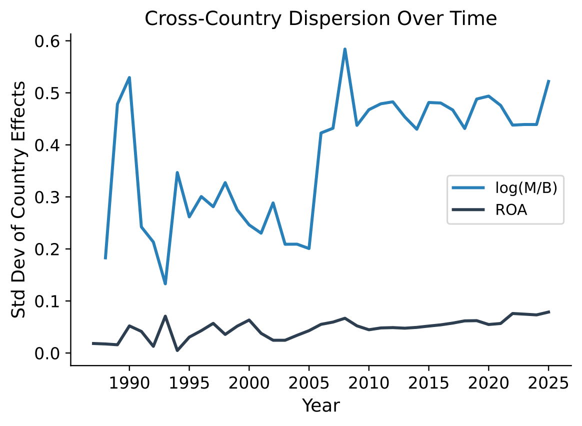
Year-by-year: $\log(M/B) \sim \text{ROA} + \log(\text{Assets}) + \text{Country FE} + \text{Industry FE}$. $\text{USA} = 0$.

Country Effects on ROA (vs USA)



Year-by-year: ROA $\sim$ log(Assets) + Country FE + Industry FE. USA = 0.

Cross-Country Dispersion Over Time



No convergence in valuations:

- M/B dispersion: 0.40 → 0.47
- Trend is **diverging**
- Spike during 2008–09 GFC

ROA dispersion more stable:

- Lower level (∼0.05–0.10)
- Spike in 2009–10 (earnings collapse)

Valuation gaps are *widening* despite globalization.

Stylized Facts

Stylized Facts (1/2): Decomposition

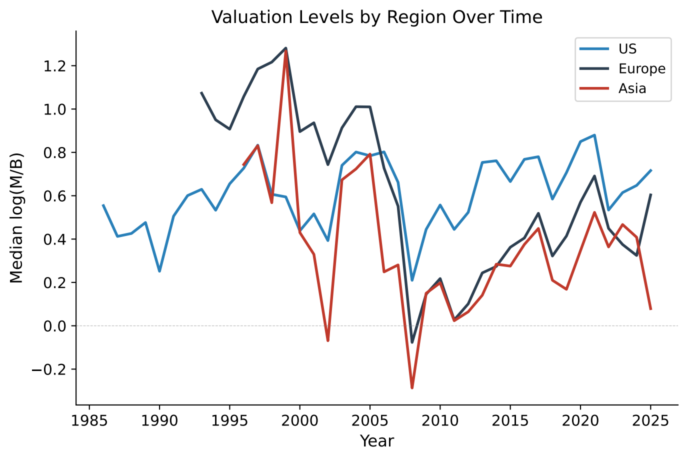
1. **Country effects dominate valuation** — country explains 50% of M/B R^2 (Shapley), vs 38% for industry. Echoes the Heston–Rouwenhorst debate, but for *levels* rather than returns.
2. **Country dominates profitability too** — for ROA, country edges industry (51% vs 43%). Both valuation and profitability are primarily location-driven.
3. **Negative M/B–ROA correlation across countries** ($\rho = -0.26$) — high-M/B countries are *not* high-ROA countries. Discount rates and growth expectations, not current profitability, drive the cross-country valuation map.

Stylized Facts (2/2): Dynamics

4. **Japan discount is persistent** — M/B effect negative in **100%** of years (avg -0.54 vs USA), with no sign of convergence.
5. **China premium collapsed post-2015** — from $+0.8$ to near zero, consistent with market correction and deleveraging.
6. **Cross-country valuation dispersion is increasing** — despite financial globalization, countries are *diverging* in M/B (std dev: $0.40 \rightarrow 0.47$).

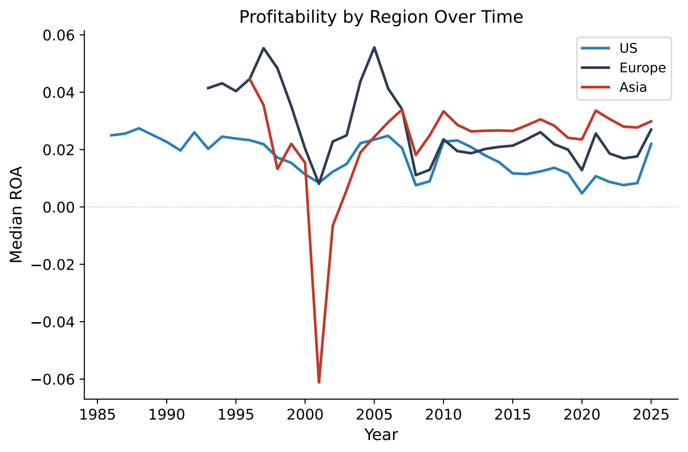
Regional Trends

Regional Valuations Over Time



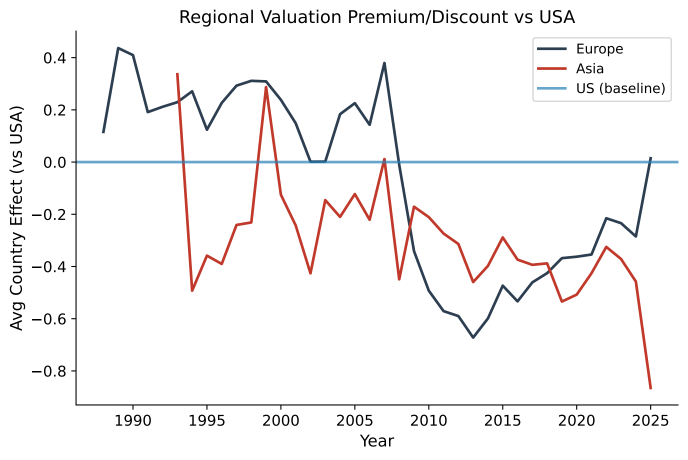
Median log(M/B) by region-year. US consistently above Europe and Asia.

Regional Profitability Over Time



Median ROA by region-year. Asia has *higher* profitability than US despite lower valuations.

Regional Valuation Gap vs USA



Average country effect (vs USA baseline) from year-by-year regressions with industry FE.

The US Valuation Premium

- US M/B is **24% higher** than Europe and **27% higher** than Asia (recent 5 years)
- Asia has *higher median ROA* than the US — yet *lower valuations*
- The gap has been **widening** since 2015, driven by US tech sector and post-GFC divergence
- Europe and Asia track each other closely, both persistently below the US baseline

Central question: Is the US premium driven by higher *growth expectations* or lower *discount rates*?

Growth vs Discount Rates

Two Hypotheses: Growth vs Discount Rates

Gordon growth model: $M/B \propto \frac{ROE-g}{r-g}$

Hypothesis 1: Growth expectations (g)

- High-M/B countries have higher expected earnings growth
- Bekaert et al. (2007, *JF*): liberalization raises growth opportunities
- Pastor–Veronesi (2003, *JF*): profitability uncertainty raises M/B

Hypothesis 2: Discount rates (r)

- Low-M/B countries have higher required returns
- Hail–Leuz (2006, *JAR*): institutions reduce cost of equity by up to 200 bps
- Stulz (2005, *JF*): expropriation risk keeps discount rates high
- Bekaert–Harvey (2000, *JF*): segmentation raises required returns

Evidence Against Pure Growth Story

- **Asia's ROA is higher than the US** — but valuations are lower. If growth tracked profitability, Asia should have *higher* M/B.
- **Negative M/B–ROA correlation** ($\rho = -0.26$): high-profitability countries are *not* high-valuation countries.
- **Japan**: strong ROA, yet deepest discount (-0.55 vs USA).
- Claus–Thomas (2001, *JF*): implied equity premia vary modestly (3–5%) across G-7.

$\Rightarrow$ Current profitability does not explain the valuation map. **Discount rates and future growth expectations** are the key drivers.

Investor-Side Explanations

Why discount rates differ across countries:

- **Investor protection**: LLSV (2002) — common-law Q premium of $\sim$0.25. Djankov et al. (2008) — anti-self-dealing index predicts valuations.
- **Market segmentation**: Bekaert–Harvey (1995, 2000) — segmented markets price local risk. Bekaert et al. (2011) — segmentation persists.
- **Cross-listing**: Doidge et al. (2004) — 17% Q premium from US governance bonding. Hail–Leuz (2009) — cost of capital falls after listing.
- **Sovereign risk**: Erb et al. (1996) — credit ratings predict expected returns.

The investor-side literature is deeper than the growth-side literature.

Literature

Related Literature: Decomposition and Institutions

Country vs industry effects:

- Heston–Rouwenhorst (1994, *JFE*) — country effects dominate for returns. We find the same for *valuation levels*.
- Griffin–Karolyi (1998, *JFE*) — industry effects matter more in traded-goods sectors.
- Brooks–Del Negro (2004) — country effects declined post-1999 for returns. Opposite for M/B.

Legal origin and investor protection:

- La Porta et al. (2002, *JF*) — investor protection raises Tobin's Q.
- Djankov et al. (2008, *JFE*) — anti-self-dealing index predicts valuations.
- Doidge et al. (2007, *JFE*) — countries matter more than firms for governance.

Related Literature: Globalization and Dynamics

Limits of convergence:

- Stulz (2005, *JF*) — "twin agency problems" prevent equalization.
- Bekaert et al. (2011, *RFS*) — segmentation decreased but persists.
- Doidge et al. (2004, *JFE*) — cross-listed firms gain $\sim$17% Q premium.

Country-specific puzzles and fundamentals:

- French–Poterba (1991, *JFE*) — Japan P/E premium; persistent discount since.
- Fama–French (1998, *JF*) — value premium exists internationally.
- Rajan–Zingales (1998, *AER*) — financial development raises valuations.

Our Contribution

- First panel decomposition of *valuation levels* (not returns) into country, industry, and time using Shapley R^2
- Largest cross-country valuation panel: 81 countries, 1986–2025, 749K firm-years
- Document **divergence** in M/B across countries, challenging the globalization-implies-equalization view
- Country M/B effects **not** explained by ROA ($\rho = -0.26$) — points to discount-rate and institutional channels